

Diachronic NLP Analysis of “Therapy-Speak” (ADHD and Autism) in General Web Discourse

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Research questions. This dissertation asks: (1) How has the prevalence of ADHD- and autism-related terminology in general web discourse changed over approximately the last decade? (2) Do these terms exhibit semantic change consistent with concept creep—in particular changes in *intensity* (vertical creep), *breadth* (horizontal creep), and *sentiment* (amelioration/degeneration)? (3) How has the thematic framing of ADHD and autism evolved over time in the public web?

Background and motivation. Research on “therapy-speak” documents the diffusion of clinical terminology into everyday discourse, raising questions about normalization, pathologization, and cultural framing of mental health (Isern-Mas and Almagro, 2025). Haslam’s concept creep framework distinguishes horizontal (widening/narrowing) and vertical (intensity shifts) semantic change in harm-related concepts (Haslam, 2016; Haslam et al., 2020). Recent computational work by Xiao et al. (2023) provides a model for empirically testing vertical creep through collocate-based severity indices in large corpora. However, existing studies focus primarily on anxiety, depression, and trauma, and rely on academic or curated general corpora. There is limited large-scale evidence on how neurodevelopmental diagnostic terms—particularly ADHD and Autism—have evolved in the broader web ecosystem. Given rising public attention, self-identification, and neurodiversity discourse, a diachronic, computational analysis of these terms in general web text is warranted. This study extends Xiao

et al.’s intensity framework by adopting the multidimensional SIBling approach to jointly examine intensity, breadth, sentiment, salience, and thematic content in large-scale web data.

Corpus and data collection strategy. The study uses Common Crawl as a large-scale archive of general web discourse. Following a pragmatic two-track design, the project will produce: (i) a *Trend Track* consisting of repeated cross-sectional yearly samples to estimate mention rates over time, and (ii) a *Corpus Track* consisting of a smaller, higher-quality collection of substantive documents for downstream NLP. To ensure temporal comparability and feasibility, one crawl per year will be selected using a consistent temporal anchor. Collection proceeds *WET-first*: Common Crawl WET files contain extracted plaintext and can be scanned efficiently to identify candidate hits and compute denominators. Candidate hits then undergo validation using their corresponding WARC records, which contain the original HTML and thus permit more effective cleaning (DOM-aware boilerplate removal), language identification, and deduplication. WARC processing is applied only to candidate/hit documents (or a controlled subset) because it is substantially more computationally expensive than WET scanning. To reduce domination by large publishers, yearly caps will be applied (e.g., ≤ 50 documents per registered domain per year). Where feasible, publication timestamps will be inferred during WARC-based validation using HTML metadata; otherwise capture time provides the primary temporal reference.

Planned analyses. The analytic plan adapts the multidimensional SIBling Framework 1.0 (Baes, 2024) and the vertical-creep testing logic of Xiao et al. (2023) to ADHD and autism in general web discourse. Three primary dimensions of lexical semantic change will be operationalised alongside prevalence and thematic evolution: (1) *Salience (prevalence)*. Using the Trend Track, yearly mention rates will be estimated as relative frequencies (candidate and validated hits per scanned corpus size), enabling a diachronic test of whether these terms are becoming more prominent in general web discourse. (2) *Intensity (vertical creep)*. Following the collocate-based approach of Xiao et al. (2023), collocates within a ± 5 -word window of the target term will be extracted per year and scored using the NRC Valence–Arousal–Dominance lexicon (Mohammad, 2025). Annual weighted arousal indices (and derived “severity” composites, where appropriate) will test whether the surrounding language becomes more or less emotionally intense over time. (3) *Sentiment*. Using the same collocate windows, annual weighted valence indices (NRC VAD) will test for shifts toward more positive vs more negative connotations (amelioration vs degeneration), reflecting potential destigmatisation or increased pejoration. (4) *Breadth (horizontal creep)*. Breadth

will be estimated via contextual dispersion: a fixed number of sentences containing the target term will be sampled per year, embedded with transformer sentence embeddings, and summarised by average pairwise cosine distance. Increasing dispersion indicates a widening range of contexts (semantic broadening). (5) *Thematic evolution*. The Corpus Track will be analysed with BERTopic (transformer embeddings, UMAP + HDBSCAN clustering, c-TF-IDF representations) to trace topics-over-time in annual bins. This inductive topic modelling approach is a deliberate departure from SIBling’s deductive, dictionary-based thematic index, and is intended to capture emergent framings (e.g., clinical vs identity/neurodiversity narratives, education/workplace accommodation, social-justice language).

Contribution. This project contributes by (i) extending concept creep and therapy-speak research to ADHD and autism, (ii) shifting evidence from platform-specific or curated corpora toward general web discourse, (iii) implementing a feasible, reproducible Common Crawl pipeline that separates trend estimation from corpus building, and (iv) refining/adapting the SIBling framework by (a) replacing Warriner norms with the larger NRC VAD lexicon for affective scoring and (b) substituting deductive thematic indices with BERTopic-based discourse “biographies”.

Feasibility and risks. Common Crawl is noisy and includes substantial boilerplate; the WET-first/WARC-validated design mitigates this by reserving expensive HTML-based cleaning for candidate hits while keeping trend denominators computationally tractable. Publication dates will be incomplete for a subset of pages; the analysis will therefore prioritise capture-year trends and treat published-time inference as opportunistic enrichment. Compute demands are manageable for an MSc through strict sampling, domain caps, and staged validation; if local throughput is insufficient, short-term rental of cloud computing resources (e.g., AWS) provides a practical scaling option.

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